

interference - practice problems

name _____

1. Two small pulses travel toward each other on the same rope. For a moment they occupy the same section of rope, and then each pulse continues on. Why is this different from two carts colliding on a track?

2. A pulse with a 0.20 m crest travels along a rope toward a pulse with a 0.15 m trough. What is the rope's displacement at the moment the crest and trough overlap?

3. Noise-canceling headphones play a sound wave that is carefully timed against the outside noise. What has to be true about the headphone wave near your ear for the outside sound to get quieter?

4. Two musicians are tuning together. Their notes are close but not exactly the same, and they hear the loudness slowly pulse louder and softer. What wave pattern are they hearing, and what should happen as the tuning improves?

5. You can hear someone talking around a hallway corner even when you cannot see them. Which wave property makes this possible, and why does it work better for sound than for a narrow beam of light?

6. A guitar string is 0.65 m long and fixed at both ends. What are the three longest wavelengths that can form standing waves on it?

7. Waves travel at 143 m/s on the guitar string from the previous problem. What are the frequencies of its first two standing waves?

8. An ambulance siren sounds higher pitched as it drives toward you and lower pitched after it passes you. What is changing at your ear, and what is the name of this effect?

9. A 1.2 m string is vibrating in a standing wave with 3 antinodes. What is the frequency of the vibration?

10. A distant galaxy has spectral lines shifted toward the red side of the spectrum. What does that redshift suggest about the galaxy's motion relative to us?

11. A child on a swing moves much higher when pushed at just the right time, even if each push is gentle. Why is this an example of resonance?

12. A student makes a standing wave on a 2.0 m rope. The wavelength is 0.50 m. How many loops (antinodes) does the standing wave have?

13. A violin's G string is 0.325 m long. When it plays its lowest standing wave, the frequency is 196 Hz. What is the wave speed on the string?